music

March 18, 2025

1 Set Up

[53]:
```
# Import
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import classification_report , f1_score
from imblearn.over_sampling import SMOTE
from sklearn.impute import SimpleImputer
from xgboost import XGBClassifier
```

[3]:
```
# Functions
def most_frequent(series):
    return series.mode()[0] if not series.mode().empty else None
```

[4]:
```
# Settings

pd.set_option('display.max_columns', None)  # Show all columns
pd.set_option('display.width', 1000)  # Prevent line breaks
pd.set_option('display.max_colwidth', None)  # Show full column values
```

[5]:
```
# Read Files

# Path to your Parquet file
parquet_file = '/Users/henryding/Documents/Seminar/Data/output_data1.parquet'

# Read the Parquet file into a DataFrame
df = pd.read_parquet(parquet_file, engine='pyarrow')  # You can also use␣
↪'fastparquet'

# Display the first few rows of the DataFrame
display(df.head(1000))
num_rows = df.shape[0]
print(f"The number of rows in the DataFrame is: {num_rows}")
```

```
   status gender     length  firstName level  lastName  registration   userId␣
 ↪            ts       auth         page  sessionId                          ␣
 ↪ location  itemInSession                                                   ␣
 ↪                                                                          ␣
 ↪userAgent                                                                  ␣
 ↪                                                               song        ␣
 ↪                                                      artist method
0       200      M   524.32934      Shlok  paid   Johnson  1.533735e+12  1749042␣
 ↪ 1538352001000  Logged In     NextSong      22683        Dallas-Fort␣
 ↪Worth-Arlington, TX            278                         "Mozilla/5.0␣
 ↪(Windows NT 6.1; WOW64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/36.0.
 ↪1985.143 Safari/537.36"                                                    ␣
 ↪                                             Ich mache einen Spiegel -␣
 ↪Dream Part 4                                                              ␣
 ↪Popol Vuh    PUT
1       200      F   238.39302    Vianney  paid    Miller  1.537500e+12  1563081␣
 ↪ 1538352002000  Logged In     NextSong      20836      San␣
 ↪Francisco-Oakland-Hayward, CA               9                 "Mozilla/5.0␣
 ↪(Macintosh; Intel Mac OS X 10_9_4) AppleWebKit/537.36 (KHTML, like Gecko)␣
 ↪Chrome/36.0.1985.143 Safari/537.36"                                        ␣
 ↪                                                                           ␣
 ↪        MiÃ Â©ntele                                                        ␣
 ↪ Los Bunkers    PUT
2       200      F   140.35546       Vina  paid    Bailey  1.536415e+12  1697168␣
 ↪ 1538352002000  Logged In     NextSong       4593                          ␣
 ↪ Hilo, HI            109                                                   ␣
 ↪   Mozilla/5.0 (Macintosh; Intel Mac OS X 10.9; rv:31.0) Gecko/20100101␣
 ↪Firefox/31.0                                                               ␣
 ↪                                                             Baby Talk     ␣
 ↪                                                           Lush    PUT
3       200      M   277.15873     Andres  paid     Foley  1.534387e+12  1222580␣
 ↪ 1538352003000  Logged In     NextSong       6370                         ␣
 ↪Watertown, SD           71                   "Mozilla/5.0 (Macintosh; Intel␣
 ↪Mac OS X 10_9_4) AppleWebKit/537.77.4 (KHTML, like Gecko) Version/7.0.5 Safari/
 ↪537.77.4"                                                                  ␣
 ↪     Horn Concerto No. 4 in E flat K495: II. Romance (Andante cantabile) ␣
 ↪Barry Tuckwell/Academy of St Martin-in-the-Fields/Sir Neville Marriner    PUT
4       200      F  1121.25342    Aaliyah  paid   Ramirez  1.537381e+12  1714398␣
 ↪ 1538352003000  Logged In     NextSong      22316          ␣
 ↪Baltimore-Columbia-Towson, MD            21                            ␣
 ↪"Mozilla/5.0 (Windows NT 6.1; WOW64) AppleWebKit/537.36 (KHTML, like Gecko)␣
 ↪Chrome/36.0.1985.143 Safari/537.36"  Close To The Edge (I. The Solid Time Of␣
 ↪Change_ II. Total Mass Retain_ III. I Get Up I Get Down_ IV. Seasons Of Man)␣
 ↪(Remastered LP Version)                                                    ␣
 ↪            Yes    PUT
```

```
..      …     …           …         …    …        …             …        …             …   ␣
 ↪    …           …        …                                           …             …   ␣
 ↪                                                                                       ␣
 ↪                                                                 …                     ␣
 ↪                                                                                       ␣
 ↪                                                 …                                     ␣
 ↪                                        …    …
995     307      F         NaN     Alivia  paid  Williams  1.535955e+12  1792538␣
 ↪ 1538352525000  Logged In    Thumbs Up       8871  New York-Newark-Jersey␣
 ↪City, NY-NJ-PA             75                                                          ␣
 ↪       Mozilla/5.0 (Macintosh; Intel Mac OS X 10.9; rv:31.0) Gecko/20100101␣
 ↪Firefox/31.0                                                                           ␣
 ↪                                                                      None             ␣
 ↪                                                               None    PUT
996     200      M   315.48036  Christian  free     Klein  1.536940e+12  1291813␣
 ↪ 1538352526000  Logged In     NextSong      14821  New York-Newark-Jersey␣
 ↪City, NY-NJ-PA             10                     "Mozilla/5.0 (Macintosh;␣
 ↪Intel Mac OS X 10_9_2) AppleWebKit/537.74.9 (KHTML, like Gecko) Version/7.0.2␣
 ↪Safari/537.74.9"                                                                       ␣
 ↪                                                                                  ␣
 ↪Odessa                                                                      Caribou␣
 ↪   PUT
997     200      M   248.73751  Kristofer  free     James  1.536879e+12  1929921␣
 ↪ 1538352526000  Logged In     NextSong       9831           ␣
 ↪Seattle-Tacoma-Bellevue, WA             56                                             ␣
 ↪                          Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:31.0)␣
 ↪Gecko/20100101 Firefox/31.0                                                            ␣
 ↪                                                                   Jigsaw␣
 ↪Falling Into Place                                                                     ␣
 ↪  Radiohead    PUT
998     200      F    53.08036       Zoey  free   Gregory  1.533190e+12  1839740␣
 ↪ 1538352527000  Logged In     NextSong      24848           ␣
 ↪Phoenix-Mesa-Scottsdale, AZ             11  "Mozilla/5.0 (X11; Linux x86_64)␣
 ↪AppleWebKit/537.36 (KHTML, like Gecko) Ubuntu Chromium/34.0.1847.116 Chrome/34.
 ↪0.1847.116 Safari/537.36"                                                              ␣
 ↪                                                     Broken Hearted Hoover Fixer␣
 ↪Sucker Guy                                                                Glen␣
 ↪Hansard    PUT
999     200      F         NaN       Zoey  free   Gregory  1.533190e+12  1839740␣
 ↪ 1538352527000  Logged In  Roll Advert      24848           ␣
 ↪Phoenix-Mesa-Scottsdale, AZ             12  "Mozilla/5.0 (X11; Linux x86_64)␣
 ↪AppleWebKit/537.36 (KHTML, like Gecko) Ubuntu Chromium/34.0.1847.116 Chrome/34.
 ↪0.1847.116 Safari/537.36"                                                              ␣
 ↪                                                                                       ␣
 ↪     None                                                                            ␣
 ↪None    GET
```

```
[1000 rows x 18 columns]
```

```
The number of rows in the DataFrame is: 26259199
```

2 Exploratory Analysis

[6]:
```
# Categorical Variables

unique_level = df['level'].unique()
print(f"level: {unique_level}")

unique_auth = df['auth'].unique()
print(f"auth: {unique_auth}")

unique_method = df['method'].unique()
print(f"method: {unique_method}")

unique_page= df['page'].unique()
print(f"page: {unique_page}")
```

```
level: ['paid' 'free']
auth: ['Logged In' 'Logged Out' 'Cancelled' 'Guest']
method: ['PUT' 'GET']
page: ['NextSong' 'Home' 'Thumbs Up' 'Error' 'Help' 'Add to Playlist'
 'Roll Advert' 'Login' 'Thumbs Down' 'Settings' 'About' 'Add Friend'
 'Upgrade' 'Submit Upgrade' 'Logout' 'Downgrade' 'Save Settings'
 'Submit Downgrade' 'Cancel' 'Cancellation Confirmation' 'Register'
 'Submit Registration']
```

[7]:
```
# User Usage
user_counts = df.groupby('userId').size().reset_index(name='count')
display(user_counts.sort_values(by='count', ascending=False))
```

	userId	count
5936	1261737	778479
12534	1564221	13591
20802	1931933	12831
163	1006695	12372
7562	1336969	11858
…	…	…
6072	1267517	1
9698	1434698	1
17778	1793623	1
9129	1408726	1
15322	1689121	1

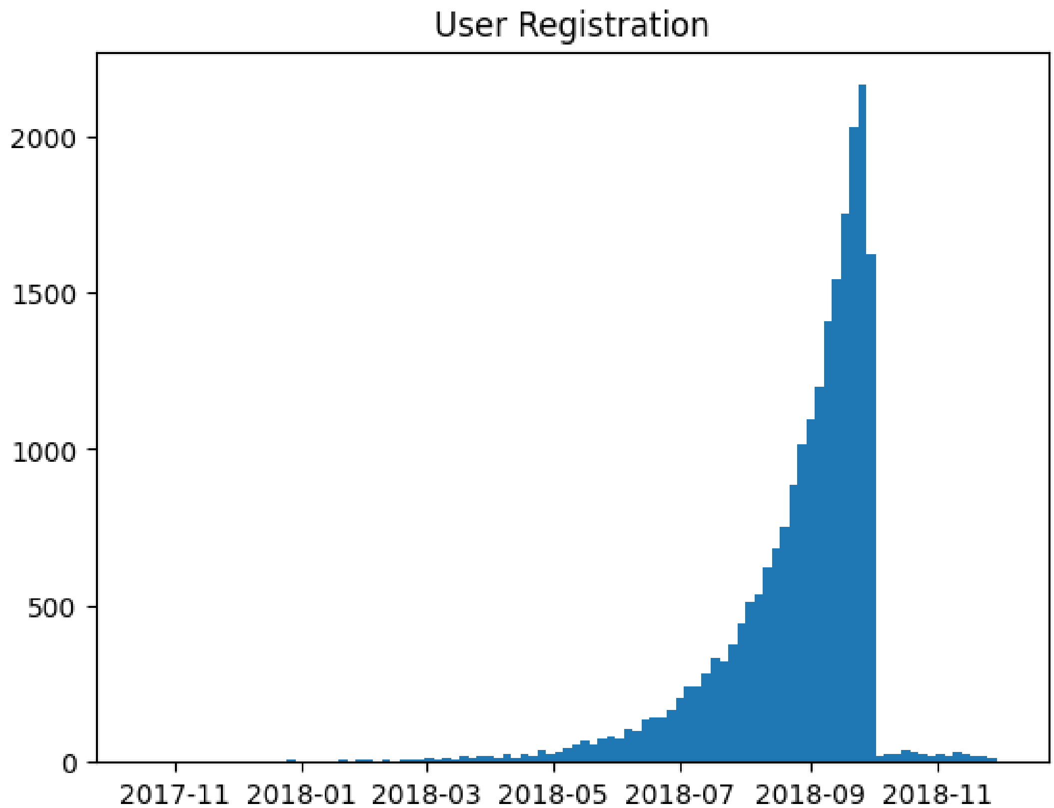
```
[22278 rows x 2 columns]
```

[8]:
```python
# Time Period
# Convert Timestamps to datetime
df['ts'] = pd.to_datetime(df['ts'], unit='ms')

print(f"timestamp min: {df['ts'].min()}")
print(f"timestamp max: {df['ts'].max()}")

# Convert Registration to datetime
df['registration'] = pd.to_datetime(df['registration'], unit='ms')

print(f"registration min: {df['registration'].min()}")
print(f"registration max: {df['registration'].max()}")
```

```
timestamp min: 2018-10-01 00:00:01
timestamp max: 2018-12-01 00:00:02
registration min: 2017-10-14 22:05:25
registration max: 2018-12-03 07:23:42
```

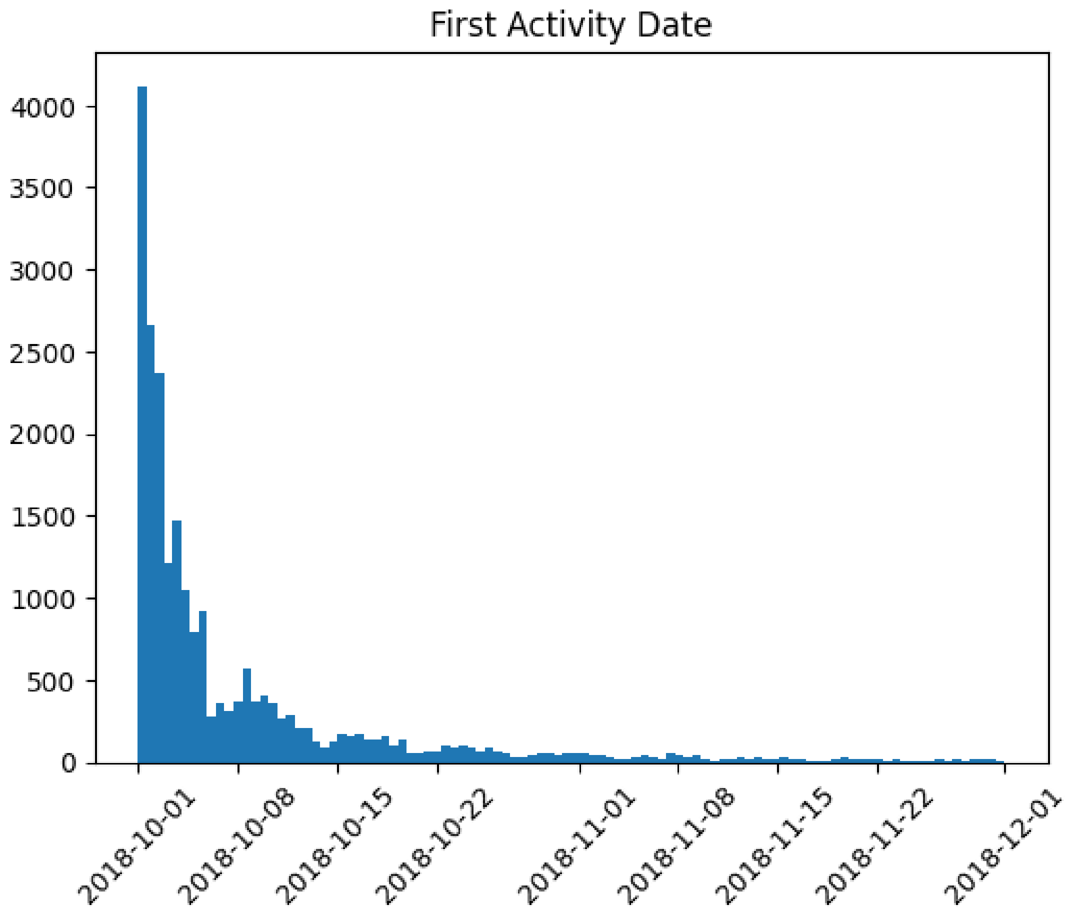
[9]:
```python
# User Registration
grouped_registration = df.groupby('userId')['registration'].agg('min')
display(grouped_registration.head())
plt.hist(grouped_registration, bins=100)
plt.title('User Registration')
plt.show()
```

```
userId
1000025   2018-07-10 09:30:08
1000035   2018-09-12 19:28:22
1000083   2018-09-07 18:01:49
1000103   2018-09-22 07:27:25
1000164   2018-08-12 09:32:01
Name: registration, dtype: datetime64[ns]
```



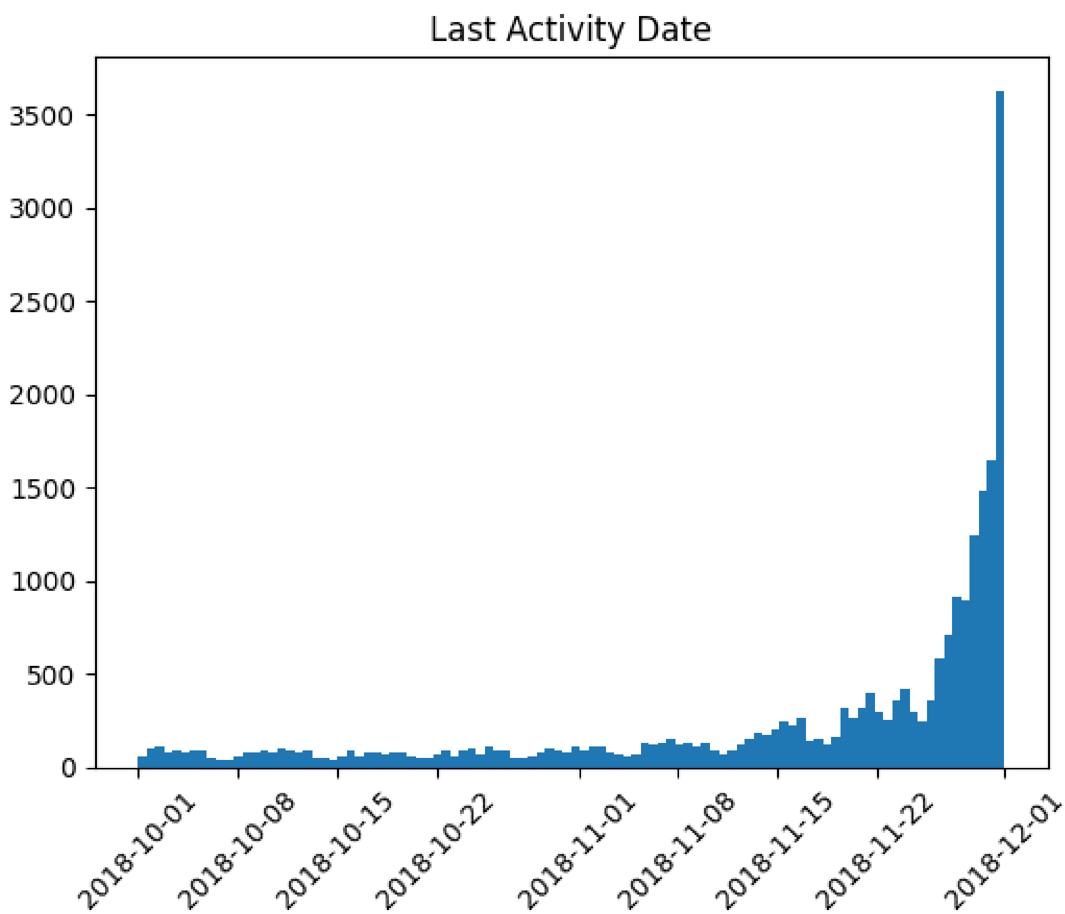
[10]:
```
# First Activity Date
grouped_start = df.groupby('userId')['ts'].agg('min')
display(grouped_start.head())
plt.hist(grouped_start, bins=100)
plt.xticks(rotation=45)
plt.title('First Activity Date')
plt.show()
```

```
userId
1000025   2018-10-02 08:59:29
1000035   2018-10-05 18:29:46
1000083   2018-10-01 05:37:45
1000103   2018-10-04 17:31:17
1000164   2018-10-01 17:40:18
Name: ts, dtype: datetime64[ns]
```



[11]:
```
# Last Activity Date
grouped_end = df.groupby('userId')['ts'].agg('max')
display(grouped_end.head())
plt.hist(grouped_end, bins=100)
plt.xticks(rotation=45)
plt.title('Last Activity Date')
plt.show()
```

```
userId
1000025   2018-10-18 20:33:05
1000035   2018-11-20 09:14:13
1000083   2018-10-12 10:04:58
1000103   2018-11-21 03:01:43
1000164   2018-11-30 16:53:06
Name: ts, dtype: datetime64[ns]
```



[12]:

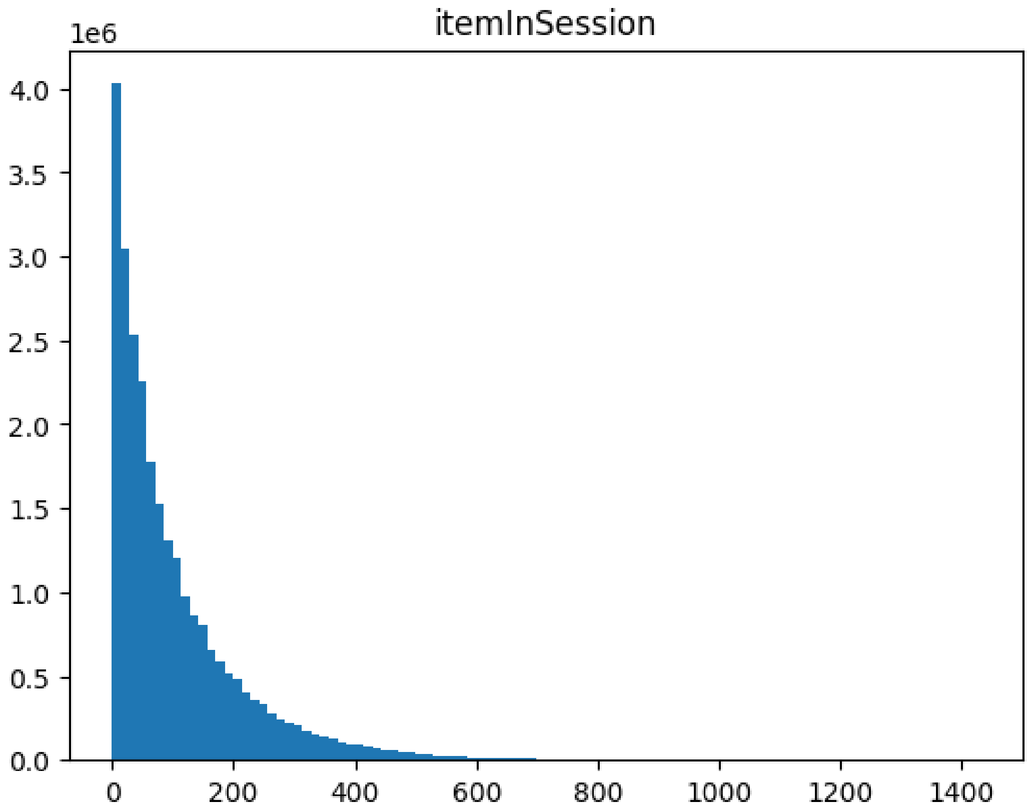
```
# High-Cardinality Variables
song_count = df.groupby('song').size().count()
print(f"Number of Songs: {song_count}")

artist_count = df.groupby('artist').size().count()
print(f"Number of Artists: {artist_count}")

user_count = df.groupby('userId').size().count()
print(f"Number of Users: {user_count}")
```

```
Number of Songs: 253564
Number of Artists: 38337
Number of Users: 22278
```

[13]:

```
# Continuous Variables

plt.hist(df['itemInSession'], bins=100)
plt.title("itemInSession")
```

```
plt.show()
```



3 Data Preprocessing

[ ]:

```

```

4 Feature Engineering

[14]:

```
# Create "week" feature to group by week
df['week'] = df['ts'].dt.to_period('W').astype('datetime64[ns]')

# Sort by userId and timestamp to calculate session_diff properly
df = df.sort_values(by=['userId', 'ts'])

# Calculate the time difference between consecutive sessions (session duration)
df['session_diff'] = df.groupby('userId')['ts'].diff().fillna(pd.
 ↪Timedelta(seconds=0))

# Calculate the total session duration (in seconds) for each session
```

```
df['session_duration'] = df['session_diff'].dt.total_seconds()

#Calculate average items per session for each user
df['session_count'] = df.groupby('userId')['itemInSession'].transform('mean')
```

[15]:
```
# Dummy Coding

# Page
for click_type in unique_page:
    df[click_type] = np.where(df['page'] == click_type, 1, 0)

# Levels
df['dummy_level'] = np.where(df['level'] == 'paid', 1, 0)
```

4.0.1 Explanation of Features:

- **week**: A new feature to group data by weeks, with each week starting on Monday.
- **session_diff**: Time difference between consecutive sessions, which helps us understand user activity over time.
- **session_duration**: Total session duration in seconds, derived from the session_diff.
- **session_count**: Number of sessions a user has in total.
- **avg_item_in_session**: Average number of items per session.
- **dummy_level**: A binary feature representing whether the user is on a free or paid plan.

[16]:
```
display(df.head())
```

```
       status gender     length firstName level lastName        registration  ␣
↪userId                     ts        auth      page  sessionId            ␣
↪location  itemInSession                                                      ␣
↪                                                 userAgent          song     ␣
↪    artist method       week    session_diff  session_duration  session_count ␣
↪NextSong  Home  Thumbs Up  Error  Help  Add to Playlist  Roll Advert  Login ␣
↪Thumbs Down  Settings  About  Add Friend  Upgrade  Submit Upgrade  Logout ␣
↪Downgrade  Save Settings  Submit Downgrade  Cancel  Cancellation Confirmation ␣
↪Register  Submit Registration  dummy_level
403570     200      M        NaN    Camren  free   Walker 2018-07-10 09:30:08 ␣
↪1000025 2018-10-02 08:59:29  Logged In       Home      23706  New␣
↪Haven-Milford, CT              0  "Mozilla/5.0 (Windows NT 6.1; WOW64)␣
↪AppleWebKit/537.36 (KHTML, like Gecko) Chrome/36.0.1985.143 Safari/537.36"   ␣
↪     None              None    GET 2018-10-01 0 days 00:00:00            0.
↪0      122.931172          0      1          0       0      0                0 ␣
↪       0       0            0         0       0            0        0        ␣
↪ 0        0           0              0                   0       0           ␣
↪          0          0                    0            0
```

```
404235     200      M  251.97669    Camren  free   Walker 2018-07-10 09:30:08 ␣
↪1000025 2018-10-02 09:02:44  Logged In  NextSong      23706  New␣
↪Haven-Milford, CT              1  "Mozilla/5.0 (Windows NT 6.1; WOW64)␣
↪AppleWebKit/537.36 (KHTML, like Gecko) Chrome/36.0.1985.143 Safari/537.36"  ␣
↪Underground                1001    PUT 2018-10-01 0 days 00:03:15           ␣
↪195.0     122.931172         1     0          0      0     0                0 ␣
↪          0      0            0         0      0           0        0        ␣
↪     0       0          0              0                0       0            ␣
↪             0         0                   0            0
405134     200      M  223.05914    Camren  free   Walker 2018-07-10 09:30:08 ␣
↪1000025 2018-10-02 09:06:55  Logged In  NextSong      23706  New␣
↪Haven-Milford, CT              2  "Mozilla/5.0 (Windows NT 6.1; WOW64)␣
↪AppleWebKit/537.36 (KHTML, like Gecko) Chrome/36.0.1985.143 Safari/537.36" ␣
↪AsesÃ Âname  Charly GarcÃ Âa     PUT 2018-10-01 0 days 00:04:11             251.
↪0      122.931172         1     0          0      0     0                0    ␣
↪      0      0            0         0      0           0        0            ␣
↪ 0       0          0              0                0       0                ␣
↪         0         0                   0            0
405929     200      M  193.38404    Camren  free   Walker 2018-07-10 09:30:08 ␣
↪1000025 2018-10-02 09:10:38  Logged In  NextSong      23706  New␣
↪Haven-Milford, CT              3  "Mozilla/5.0 (Windows NT 6.1; WOW64)␣
↪AppleWebKit/537.36 (KHTML, like Gecko) Chrome/36.0.1985.143 Safari/537.36"    ␣
↪Last Nite       The Strokes    PUT 2018-10-01 0 days 00:03:43             223.
↪0      122.931172         1     0          0      0     0                0    ␣
↪      0      0            0         0      0           0        0            ␣
↪ 0       0          0              0                0       0                ␣
↪         0         0                   0            0
406597     200      M  177.37098    Camren  free   Walker 2018-07-10 09:30:08 ␣
↪1000025 2018-10-02 09:13:51  Logged In  NextSong      23706  New␣
↪Haven-Milford, CT              4  "Mozilla/5.0 (Windows NT 6.1; WOW64)␣
↪AppleWebKit/537.36 (KHTML, like Gecko) Chrome/36.0.1985.143 Safari/537.36"    ␣
↪  The End          Pearl Jam    PUT 2018-10-01 0 days 00:03:13             193.
↪0      122.931172         1     0          0      0     0                0    ␣
↪      0      0            0         0      0           0        0            ␣
↪ 0       0          0              0                0       0                ␣
↪         0         0                   0            0
```

5 Defining Churn and Activity

[17]:
```
# Identify users who have data for the current week (check for missing rows for␣
↪each user)
all_users = df['userId'].unique()
weeks_in_df = df['week'].unique()

# Create a DataFrame of all possible combinations of userId and week (even for␣
↪missing users)
```

```
all_combinations = pd.MultiIndex.from_product([all_users, weeks_in_df],␣
↪names=['userId', 'week'])
df_all_weeks = pd.DataFrame(index=all_combinations).reset_index()

# Merge with the original data to indentify users missing for a given week
df_all_weeks = df_all_weeks.merge(df, on=['userId', 'week'], how='left')
df_all_weeks['session_count'] = df_all_weeks['session_count'].fillna(0)

# Fill missing session_count values with 0 (since missing weeks for users)

# Sort by userId and week to handle previous week properly
df_all_weeks = df_all_weeks.sort_values(by=['userId', 'week'])

# Create "prev_week_activity" to track if the user was active in the previous␣
↪week
df_all_weeks['prev_week_activity'] = df_all_weeks.groupby('userId',␣
↪group_keys=False)['session_count'].shift(1).fillna(0)

# Define churn: users who were active in the previous week but are inactive now
df_all_weeks['churn'] = np.where(
    ((df_all_weeks['session_count'] == 0) & (df_all_weeks['prev_week_activity']␣
↪> 0)),
    1,
    0
)
```

[18]:
```
df_all_weeks = df_all_weeks.sort_values(by='churn', ascending=False)
display(df_all_weeks.head(100))

print(df_all_weeks['churn'].dtypes)
print(df_all_weeks['churn'].unique())
```

```
          userId       week  status gender  length firstName level lastName␣
↪registration  ts auth page  sessionId location  itemInSession userAgent song␣
↪artist method session_diff  session_duration  session_count  NextSong  Home ␣
↪Thumbs Up  Error  Help  Add to Playlist  Roll Advert  Login  Thumbs Down ␣
↪Settings  About  Add Friend  Upgrade  Submit Upgrade  Logout  Downgrade  Save␣
↪Settings  Submit Downgrade  Cancel  Cancellation Confirmation  Register ␣
↪Submit Registration  dummy_level  prev_week_activity  churn
14614214  1537354 2018-11-26     NaN     NaN     NaN       NaN   NaN      NaN    ␣
↪      NaT NaT  NaN  NaN         NaN       NaN              NaN        NaN  NaN    ␣
↪NaN     NaN           NaT                 NaN            0.0        NaN    NaN       ␣
↪ NaN     NaN    NaN                 NaN          NaN     NaN             NaN       NaN ␣
↪  NaN          NaN       NaN               NaN      NaN         NaN              NaN  ␣
↪            NaN      NaN                         NaN        NaN                    ␣
↪NaN          NaN           84.900563      1
```

```
22391406  1840707 2018-10-15     NaN     NaN      NaN        NaN    NaN       NaN
↪      NaT NaT  NaN  NaN         NaN       NaN             NaN        NaN  NaN
↪NaN     NaN          NaT                NaN            0.0        NaN    NaN
↪ NaN     NaN    NaN                NaN          NaN     NaN            NaN         NaN
↪  NaN           NaN       NaN               NaN      NaN         NaN              NaN
↪             NaN      NaN                        NaN         NaN
↪NaN            NaN           21.700000       1
26197262  1994416 2018-11-19     NaN     NaN      NaN        NaN    NaN       NaN
↪      NaT NaT  NaN  NaN         NaN       NaN             NaN        NaN  NaN
↪NaN     NaN          NaT                NaN            0.0        NaN    NaN
↪ NaN     NaN    NaN                NaN          NaN     NaN            NaN         NaN
↪  NaN           NaN       NaN               NaN      NaN         NaN              NaN
↪             NaN      NaN                        NaN         NaN
↪NaN            NaN           14.100000       1
22700641  1854015 2018-10-22     NaN     NaN      NaN        NaN    NaN       NaN
↪      NaT NaT  NaN  NaN         NaN       NaN             NaN        NaN  NaN
↪NaN     NaN          NaT                NaN            0.0        NaN    NaN
↪ NaN     NaN    NaN                NaN          NaN     NaN            NaN         NaN
↪  NaN           NaN       NaN               NaN      NaN         NaN              NaN
↪             NaN      NaN                        NaN         NaN
↪NaN            NaN            4.368421       1
16133336  1596750 2018-10-15     NaN     NaN      NaN        NaN    NaN       NaN
↪      NaT NaT  NaN  NaN         NaN       NaN             NaN        NaN  NaN
↪NaN     NaN          NaT                NaN            0.0        NaN    NaN
↪ NaN     NaN    NaN                NaN          NaN     NaN            NaN         NaN
↪  NaN           NaN       NaN               NaN      NaN         NaN              NaN
↪             NaN      NaN                        NaN         NaN
↪NaN            NaN           53.375000       1
…                 …         …      …     …      …       …    …       …          …   ..  …   …
↪       …      …            …      …  …    …    …          …              …
↪         …       …   …       …    …   …                …          …    …
↪ …       …    …         …      …             …     …        …            …
↪         …     …                         …       …                  …
↪…                   …    …
4854998   1185454 2018-11-05     NaN     NaN      NaN        NaN    NaN       NaN
↪      NaT NaT  NaN  NaN         NaN       NaN             NaN        NaN  NaN
↪NaN     NaN          NaT                NaN            0.0        NaN    NaN
↪ NaN     NaN    NaN                NaN          NaN     NaN            NaN         NaN
↪  NaN           NaN       NaN               NaN      NaN         NaN              NaN
↪             NaN      NaN                        NaN         NaN
↪NaN            NaN           67.903955       1
```

```
15366635  1566527 2018-10-29     NaN     NaN      NaN        NaN    NaN       NaN    ␣
↪      NaT NaT  NaN  NaN         NaN       NaN             NaN       NaN  NaN  ␣
↪NaN     NaN           NaT               NaN            0.0        NaN   NaN      ␣
↪ NaN     NaN   NaN                NaN          NaN    NaN          NaN       NaN ␣
↪  NaN          NaN       NaN              NaN      NaN         NaN            NaN ␣
↪            NaN      NaN                       NaN       NaN                    ␣
↪NaN           NaN          130.575124       1
18389900  1683635 2018-11-12     NaN     NaN      NaN        NaN    NaN       NaN    ␣
↪      NaT NaT  NaN  NaN         NaN       NaN             NaN       NaN  NaN  ␣
↪NaN     NaN           NaT               NaN            0.0        NaN   NaN      ␣
↪ NaN     NaN   NaN                NaN          NaN    NaN          NaN       NaN ␣
↪  NaN          NaN       NaN              NaN      NaN         NaN            NaN ␣
↪            NaN      NaN                       NaN       NaN                    ␣
↪NaN           NaN           19.950000       1
23071750  1869407 2018-10-15     NaN     NaN      NaN        NaN    NaN       NaN    ␣
↪      NaT NaT  NaN  NaN         NaN       NaN             NaN       NaN  NaN  ␣
↪NaN     NaN           NaT               NaN            0.0        NaN   NaN      ␣
↪ NaN     NaN   NaN                NaN          NaN    NaN          NaN       NaN ␣
↪  NaN          NaN       NaN              NaN      NaN         NaN            NaN ␣
↪            NaN      NaN                       NaN       NaN                    ␣
↪NaN           NaN           17.458763       1
6504769   1246937 2018-10-08     NaN     NaN      NaN        NaN    NaN       NaN    ␣
↪      NaT NaT  NaN  NaN         NaN       NaN             NaN       NaN  NaN  ␣
↪NaN     NaN           NaT               NaN            0.0        NaN   NaN      ␣
↪ NaN     NaN   NaN                NaN          NaN    NaN          NaN       NaN ␣
↪  NaN          NaN       NaN              NaN      NaN         NaN            NaN ␣
↪            NaN      NaN                       NaN       NaN                    ␣
↪NaN           NaN           35.305970       1

[100 rows x 47 columns]

int64
[1 0]
```

[19]:
```
display(df_all_weeks[df_all_weeks['userId'] == 1566527].sort_values('week'))
```

```
          userId       week  status gender     length firstName level lastName ␣
↪      registration                    ts        auth       page  sessionId  ␣
↪location  itemInSession                                                        ␣
↪                                                         userAgent              ␣
↪                    song                                  artist method  ␣
↪session_diff  session_duration  session_count  NextSong  Home  Thumbs Up ␣
↪Error  Help  Add to Playlist  Roll Advert  Login  Thumbs Down  Settings  About␣
↪ Add Friend  Upgrade  Submit Upgrade  Logout  Downgrade  Save Settings  Submit␣
↪Downgrade  Cancel  Cancellation Confirmation  Register  Submit Registration ␣
↪dummy_level  prev_week_activity  churn
```

```
15365629  1566527 2018-10-01     NaN     NaN         NaN         NaN    NaN       NaN ␣
↪                NaT                  NaT         NaN         NaN         NaN          ␣
↪NaN            NaN                                                                    ␣
↪                                                NaN                                   ␣
↪              NaN                                      NaN     NaN                   ␣
↪NaT               NaN        0.000000         NaN    NaN         NaN     NaN    NaN  ␣
↪            NaN          NaN     NaN           NaN         NaN     NaN          NaN  ␣
↪   NaN               NaN      NaN          NaN             NaN                NaN    ␣
↪NaN                           NaN        NaN                  NaN            NaN     ␣
↪       0.000000       0
15365703  1566527 2018-10-08   200.0       F         NaN    Stella  paid      Lee␣
↪2018-09-26 13:58:12 2018-10-12 23:11:28  Logged In      Home     13763.0 ␣
↪Odessa, TX            77.0  "Mozilla/5.0 (Windows NT 6.1; WOW64) AppleWebKit/
↪537.36 (KHTML, like Gecko) Chrome/36.0.1985.143 Safari/537.36"                  ␣
↪                       None                                       None    GET 0␣
↪days 00:00:03              3.0      130.575124        0.0   1.0         0.0    0.
↪0   0.0               0.0          0.0     0.0           0.0        0.0     0.0    ␣
↪    0.0       0.0              0.0      0.0         0.0            0.0            ␣
↪  0.0      0.0                        0.0        0.0                  0.0         ␣
↪  1.0          130.575124      0
15365632  1566527 2018-10-08   200.0       F  197.79873    Stella  paid      Lee␣
↪2018-09-26 13:58:12 2018-10-12 19:15:21  Logged In  NextSong    13763.0 ␣
↪Odessa, TX              2.0  "Mozilla/5.0 (Windows NT 6.1; WOW64) AppleWebKit/
↪537.36 (KHTML, like Gecko) Chrome/36.0.1985.143 Safari/537.36"                  ␣
↪                SÃ Â© Morir                         AndrÃ Â©s Cepeda    PUT 0␣
↪days 00:03:31            211.0      130.575124        1.0   0.0         0.0    0.
↪0   0.0               0.0          0.0     0.0           0.0        0.0     0.0    ␣
↪    0.0       0.0              0.0      0.0         0.0            0.0            ␣
↪  0.0      0.0                        0.0        0.0                  0.0         ␣
↪  1.0          130.575124      0
15365633  1566527 2018-10-08   200.0       F  301.32200    Stella  paid      Lee␣
↪2018-09-26 13:58:12 2018-10-12 19:18:38  Logged In  NextSong    13763.0 ␣
↪Odessa, TX              3.0  "Mozilla/5.0 (Windows NT 6.1; WOW64) AppleWebKit/
↪537.36 (KHTML, like Gecko) Chrome/36.0.1985.143 Safari/537.36"  Just The Two␣
↪Of Us (WSM Compilation Edit)  Grover Washington Jr. feat. Bill Withers    PUT␣
↪0 days 00:03:17            197.0      130.575124        1.0   0.0         0.0    ␣
↪0.0   0.0               0.0          0.0     0.0           0.0        0.0     0.0  ␣
↪      0.0       0.0              0.0      0.0         0.0            0.0          ␣
↪    0.0      0.0                        0.0        0.0                  0.0       ␣
↪    1.0          130.575124      0
```

```
15365634  1566527 2018-10-08   200.0      F  214.07302    Stella  paid      Lee
 ↪2018-09-26 13:58:12 2018-10-12 19:23:39  Logged In  NextSong    13763.0
 ↪Odessa, TX            4.0  "Mozilla/5.0 (Windows NT 6.1; WOW64) AppleWebKit/
 ↪537.36 (KHTML, like Gecko) Chrome/36.0.1985.143 Safari/537.36"
 ↪                    Shimmer                                        Fuel    PUT 0
 ↪days 00:05:01             301.0      130.575124       1.0    0.0         0.0    0.
 ↪0    0.0              0.0           0.0     0.0           0.0        0.0     0.0
 ↪    0.0       0.0            0.0      0.0         0.0            0.0
 ↪  0.0      0.0                        0.0        0.0                 0.0
 ↪  1.0          130.575124       0
…            …        …     …    …        …       …   …       …                 …
 ↪                …        …       …        …         …            …
 ↪
 ↪                  …                                          …
 ↪                   …     …             …                …            …       …
 ↪…        …    …   …              …          …    …          …       …    …
 ↪    …      …             …     …        …            …               …     …
 ↪                   …       …                  …          …                  …
 ↪   …
15366635  1566527 2018-10-29     NaN     NaN         NaN        NaN   NaN       NaN
 ↪               NaT                 NaT         NaN        NaN         NaN
 ↪NaN            NaN
 ↪                                                  NaN
 ↪             NaN                                        NaN     NaN
 ↪NaT                NaN        0.000000         NaN    NaN         NaN     NaN   NaN
 ↪           NaN           NaN     NaN            NaN        NaN     NaN         NaN
 ↪   NaN              NaN      NaN          NaN            NaN                NaN
 ↪NaN                        NaN         NaN                NaN            NaN
 ↪     130.575124        1
15366636  1566527 2018-11-05     NaN     NaN         NaN        NaN   NaN       NaN
 ↪               NaT                 NaT         NaN        NaN         NaN
 ↪NaN            NaN
 ↪                                                  NaN
 ↪             NaN                                        NaN     NaN
 ↪NaT                NaN        0.000000         NaN    NaN         NaN     NaN   NaN
 ↪           NaN           NaN     NaN            NaN        NaN     NaN         NaN
 ↪   NaN              NaN      NaN          NaN            NaN                NaN
 ↪NaN                        NaN         NaN                NaN            NaN
 ↪       0.000000       0
```

```
15366637  1566527 2018-11-12     NaN     NaN         NaN        NaN    NaN       NaN ␣
↪                 NaT                   NaT         NaN        NaN         NaN      ␣
↪NaN              NaN                                                               ␣
↪                                                NaN                                ␣
↪               NaN                                   NaN     NaN                   ␣
↪NaT                NaN         0.000000        NaN    NaN         NaN     NaN    NaN ␣
↪             NaN            NaN     NaN             NaN        NaN     NaN        NaN ␣
↪   NaN              NaN      NaN         NaN            NaN                NaN     ␣
↪NaN                          NaN        NaN                   NaN          NaN     ␣
↪       0.000000       0
15366638  1566527 2018-11-19     NaN     NaN         NaN        NaN    NaN       NaN ␣
↪                 NaT                   NaT         NaN        NaN         NaN      ␣
↪NaN              NaN                                                               ␣
↪                                                NaN                                ␣
↪               NaN                                   NaN     NaN                   ␣
↪NaT                NaN         0.000000        NaN    NaN         NaN     NaN    NaN ␣
↪             NaN            NaN     NaN             NaN        NaN     NaN        NaN ␣
↪   NaN              NaN      NaN         NaN            NaN                NaN     ␣
↪NaN                          NaN        NaN                   NaN          NaN     ␣
↪       0.000000       0
15366639  1566527 2018-11-26     NaN     NaN         NaN        NaN    NaN       NaN ␣
↪                 NaT                   NaT         NaN        NaN         NaN      ␣
↪NaN              NaN                                                               ␣
↪                                                NaN                                ␣
↪               NaN                                   NaN     NaN                   ␣
↪NaT                NaN         0.000000        NaN    NaN         NaN     NaN    NaN ␣
↪             NaN            NaN     NaN             NaN        NaN     NaN        NaN ␣
↪   NaN              NaN      NaN         NaN            NaN                NaN     ␣
↪NaN                          NaN        NaN                   NaN          NaN     ␣
↪       0.000000       0

[1011 rows x 47 columns]
```

[30]:
```
# Testing purposes:

df_first_weeks = df_all_weeks[(df_all_weeks['week'] == '2018-10-01') |␣
 ↪(df_all_weeks['week'] == '2018-10-08')]
display(df_first_weeks)
```

```
           userId       week  status gender     length firstName level lastName ␣
↪      registration                  ts        auth                page  sessionId ␣
↪             location  itemInSession                                               ␣
↪                      userAgent                                     song           ␣
↪          artist method    session_diff  session_duration  session_count ␣
↪NextSong  Home  Thumbs Up  Error  Help  Add to Playlist  Roll Advert  Login ␣
↪Thumbs Down  Settings  About  Add Friend  Upgrade  Submit Upgrade  Logout ␣
↪Downgrade  Save Settings  Submit Downgrade  Cancel  Cancellation Confirmation ␣
↪Register  Submit Registration  dummy_level  prev_week_activity  churn
```

```
23374326  1881831 2018-10-08     NaN     NaN         NaN        NaN   NaN        NaN ␣
↪                NaT                   NaT         NaN                NaN         NaN ␣
↪                  NaN            NaN                                                 ␣
↪                            NaN                                  NaN                 ␣
↪             NaN     NaN              NaT                NaN        0.000000        ␣
↪NaN    NaN        NaN     NaN    NaN                NaN          NaN     NaN         ␣
↪ NaN        NaN     NaN          NaN       NaN              NaN       NaN        NaN ␣
↪           NaN                NaN      NaN                         NaN         NaN  ␣
↪                NaN            NaN           55.924138       1
923587    1036445 2018-10-08     NaN     NaN         NaN        NaN   NaN        NaN ␣
↪                NaT                   NaT         NaN                NaN         NaN ␣
↪                  NaN            NaN                                                 ␣
↪                            NaN                                  NaN                 ␣
↪             NaN     NaN              NaT                NaN        0.000000        ␣
↪NaN    NaN        NaN     NaN    NaN                NaN          NaN     NaN         ␣
↪ NaN        NaN     NaN          NaN       NaN              NaN       NaN        NaN ␣
↪           NaN                NaN      NaN                         NaN         NaN  ␣
↪                NaN            NaN           77.488127       1
11904905  1428147 2018-10-08     NaN     NaN         NaN        NaN   NaN        NaN ␣
↪                NaT                   NaT         NaN                NaN         NaN ␣
↪                  NaN            NaN                                                 ␣
↪                            NaN                                  NaN                 ␣
↪             NaN     NaN              NaT                NaN        0.000000        ␣
↪NaN    NaN        NaN     NaN    NaN                NaN          NaN     NaN         ␣
↪ NaN        NaN     NaN          NaN       NaN              NaN       NaN        NaN ␣
↪           NaN                NaN      NaN                         NaN         NaN  ␣
↪                NaN            NaN           29.664634       1
18389805  1683635 2018-10-08     NaN     NaN         NaN        NaN   NaN        NaN ␣
↪                NaT                   NaT         NaN                NaN         NaN ␣
↪                  NaN            NaN                                                 ␣
↪                            NaN                                  NaN                 ␣
↪             NaN     NaN              NaT                NaN        0.000000        ␣
↪NaN    NaN        NaN     NaN    NaN                NaN          NaN     NaN         ␣
↪ NaN        NaN     NaN          NaN       NaN              NaN       NaN        NaN ␣
↪           NaN                NaN      NaN                         NaN         NaN  ␣
↪                NaN            NaN           19.950000       1
35485     1001177 2018-10-08     NaN     NaN         NaN        NaN   NaN        NaN ␣
↪                NaT                   NaT         NaN                NaN         NaN ␣
↪                  NaN            NaN                                                 ␣
↪                            NaN                                  NaN                 ␣
↪             NaN     NaN              NaT                NaN        0.000000        ␣
↪NaN    NaN        NaN     NaN    NaN                NaN          NaN     NaN         ␣
↪ NaN        NaN     NaN          NaN       NaN              NaN       NaN        NaN ␣
↪           NaN                NaN      NaN                         NaN         NaN  ␣
↪                NaN            NaN           49.925187       1
```

```
...              ...        ...      ...      ...          ...        ...    ...        ...                   ...
                      ...        ...                ...        ...                         ...
    ...                                                                                           ...
                                          ...                                        ...      ...             ...
              ...              ...        ...    ...          ...      ...    ...                ...
  ...     ...            ...          ...     ...          ...        ...               ...       ...         ...
             ...                  ...     ...                                ...         ...
      ...            ...                      ...     ...
8783867   1307769 2018-10-01   200.0      M  431.62077     Josue  free  Pearson
2018-06-28 13:37:39 2018-10-05 18:04:14  Logged In         NextSong    12922.0
 Sault Ste. Marie, MI            7.0  Mozilla/5.0 (Windows NT 6.2; WOW64; rv:
31.0) Gecko/20100101 Firefox/31.0  The Saga Of Harrison Crabfeathers  Mike
Marshall & Chris Thile     PUT 0 days 00:04:50             290.0       53.899563
       1.0    0.0         0.0     0.0    0.0              0.0           0.0     0.0
        0.0        0.0     0.0          0.0       0.0              0.0      0.0
 0.0            0.0              0.0      0.0                          0.0
0.0                  0.0            0.0           53.899563      0
8783866   1307769 2018-10-01   200.0      M  290.24608     Josue  free  Pearson
2018-06-28 13:37:39 2018-10-05 17:59:24  Logged In         NextSong    12922.0
 Sault Ste. Marie, MI            6.0  Mozilla/5.0 (Windows NT 6.2; WOW64; rv:
31.0) Gecko/20100101 Firefox/31.0                                  Hook
     Blues Traveler     PUT 0 days 00:02:52             172.0       53.899563
   1.0    0.0         0.0     0.0    0.0              0.0           0.0     0.0
    0.0        0.0     0.0          0.0       0.0              0.0      0.0        0.
0            0.0              0.0      0.0                          0.0       0.
0                  0.0            0.0           53.899563      0
8783865   1307769 2018-10-01   200.0      M        NaN     Josue  free  Pearson
2018-06-28 13:37:39 2018-10-05 17:56:32  Logged In      Roll Advert    12922.0
 Sault Ste. Marie, MI            5.0  Mozilla/5.0 (Windows NT 6.2; WOW64; rv:
31.0) Gecko/20100101 Firefox/31.0                                  None
                None     GET 0 days 00:03:08             188.0       53.899563
   0.0    0.0         0.0     0.0    0.0              0.0           1.0     0.0
    0.0        0.0     0.0          0.0       0.0              0.0      0.0        0.
0            0.0              0.0      0.0                          0.0       0.
0                  0.0            0.0           53.899563      0
8783864   1307769 2018-10-01   200.0      M  360.51546     Josue  free  Pearson
2018-06-28 13:37:39 2018-10-05 17:53:24  Logged In         NextSong    12922.0
 Sault Ste. Marie, MI            4.0  Mozilla/5.0 (Windows NT 6.2; WOW64; rv:
31.0) Gecko/20100101 Firefox/31.0                             Crumpshit
   Philippe Rochard     PUT 0 days 00:01:47             107.0       53.899563
   1.0    0.0         0.0     0.0    0.0              0.0           0.0     0.0
    0.0        0.0     0.0          0.0       0.0              0.0      0.0        0.
0            0.0              0.0      0.0                          0.0       0.
0                  0.0            0.0           53.899563      0
```

```
8783863   1307769 2018-10-01   200.0      M        NaN     Josue  free  Pearson␣
↪2018-06-28 13:37:39 2018-10-05 17:51:37  Logged In  Add to Playlist    12922.0␣
↪ Sault Ste. Marie, MI            3.0  Mozilla/5.0 (Windows NT 6.2; WOW64; rv:
↪31.0) Gecko/20100101 Firefox/31.0                               None          ␣
↪             None    PUT 0 days 00:02:48            168.0      53.899563      ␣
↪  0.0   0.0         0.0     0.0   0.0               1.0          0.0     0.0   ␣
↪   0.0        0.0     0.0         0.0       0.0            0.0      0.0       0.
↪0            0.0               0.0      0.0                      0.0       0.
↪0                  0.0          0.0          53.899563      0

[6264368 rows x 47 columns]

           userId       week  status gender     length firstName level lastName ␣
↪      registration                  ts       auth         page  sessionId      ␣
↪                           location  itemInSession                             ␣
↪                                    userAgent                                  ␣
↪    song           artist method    session_diff  session_duration ␣
↪session_count  NextSong  Home  Thumbs Up  Error  Help  Add to Playlist  Roll␣
↪Advert  Login  Thumbs Down  Settings  About  Add Friend  Upgrade  Submit␣
↪Upgrade  Logout  Downgrade  Save Settings  Submit Downgrade  Cancel ␣
↪Cancellation Confirmation  Register  Submit Registration  dummy_level ␣
↪prev_week_activity  churn
23374326  1881831 2018-10-08     NaN    NaN        NaN       NaN   NaN      NaN ␣
↪              NaT                 NaT        NaN          NaN         NaN      ␣
↪                               NaN            NaN                              ␣
↪                                         NaN                                   ␣
↪      NaN              NaN    NaN              NaT               NaN        0.
↪000000        NaN   NaN         NaN    NaN   NaN              NaN          NaN ␣
↪ NaN          NaN        NaN    NaN         NaN       NaN             NaN    ␣
↪NaN         NaN            NaN              NaN     NaN                        ␣
↪NaN        NaN                  NaN          NaN          55.924138       1
923587    1036445 2018-10-08     NaN    NaN        NaN       NaN   NaN      NaN ␣
↪              NaT                 NaT        NaN          NaN         NaN      ␣
↪                               NaN            NaN                              ␣
↪                                         NaN                                   ␣
↪      NaN              NaN    NaN              NaT               NaN        0.
↪000000        NaN   NaN         NaN    NaN   NaN              NaN          NaN ␣
↪ NaN          NaN        NaN    NaN         NaN       NaN             NaN    ␣
↪NaN         NaN            NaN              NaN     NaN                        ␣
↪NaN        NaN                  NaN          NaN          77.488127       1
```

```
11904905  1428147 2018-10-08     NaN    NaN         NaN       NaN   NaN       NaN ␣
↪                NaT                  NaT        NaN          NaN        NaN     ␣
↪                                NaN           NaN                                ␣
↪                                          NaN                                    ␣
↪     NaN              NaN    NaN              NaT               NaN       0.
↪000000        NaN   NaN         NaN    NaN   NaN               NaN          NaN  ␣
↪ NaN          NaN        NaN    NaN         NaN       NaN              NaN    ␣
↪NaN         NaN            NaN                NaN     NaN                        ␣
↪NaN        NaN                  NaN           NaN           29.664634      1
18389805  1683635 2018-10-08     NaN    NaN         NaN       NaN   NaN       NaN ␣
↪                NaT                  NaT        NaN          NaN        NaN     ␣
↪                                NaN           NaN                                ␣
↪                                          NaN                                    ␣
↪     NaN              NaN    NaN              NaT               NaN       0.
↪000000        NaN   NaN         NaN    NaN   NaN               NaN          NaN  ␣
↪ NaN          NaN        NaN    NaN         NaN       NaN              NaN    ␣
↪NaN         NaN            NaN                NaN     NaN                        ␣
↪NaN        NaN                  NaN           NaN           19.950000      1
35485     1001177 2018-10-08     NaN    NaN         NaN       NaN   NaN       NaN ␣
↪                NaT                  NaT        NaN          NaN        NaN     ␣
↪                                NaN           NaN                                ␣
↪                                          NaN                                    ␣
↪     NaN              NaN    NaN              NaT               NaN       0.
↪000000        NaN   NaN         NaN    NaN   NaN               NaN          NaN  ␣
↪ NaN          NaN        NaN    NaN         NaN       NaN              NaN    ␣
↪NaN         NaN            NaN                NaN     NaN                        ␣
↪NaN        NaN                  NaN           NaN           49.925187      1
…           …        …     …    …        …       …   …      …                 … ␣
↪                …        …          …        …                               ␣
↪     …            …                                                          ␣
↪           …                                   …             …    …          ␣
↪      …             …            …       …   …        …    …   …             ␣
↪…          …    …         …       …    …        …       …             …     …␣
↪       …            …               …     …                        …      …  ␣
↪                …          …                 …    …
8782210   1307763 2018-10-08   200.0       F  166.00771     Khloe  free    Moore␣
↪2018-08-11 20:41:52 2018-10-11 16:08:24  Logged In     NextSong    61487.0 ␣
↪Sacramento--Roseville--Arden-Arcade, CA          96.0  Mozilla/5.0 (Windows␣
↪NT 6.1; WOW64; rv:31.0) Gecko/20100101 Firefox/31.0                          ␣
↪    The Middle  Jimmy Eat World    PUT 0 days 00:06:16            376.0     ␣
↪125.962733       1.0    0.0         0.0     0.0    0.0              0.0          0.
↪0     0.0           0.0        0.0     0.0          0.0       0.0             0.0  ␣
↪ 0.0        0.0            0.0                0.0     0.0                        ␣
↪ 0.0       0.0                   0.0          0.0           125.962733      0
```

```
8782209   1307763 2018-10-08   200.0      F  376.71138     Khloe  free    Moore␣
 ↪2018-08-11 20:41:52 2018-10-11 16:02:08  Logged In     NextSong    61487.0 ␣
 ↪Sacramento--Roseville--Arden-Arcade, CA           95.0  Mozilla/5.0 (Windows␣
 ↪NT 6.1; WOW64; rv:31.0) Gecko/20100101 Firefox/31.0  Another Rainbow␣
 ↪(HardAct2Follow Clubmix)  Magnus Carlsson    PUT 0 days 00:03:53           ␣
 ↪233.0     125.962733       1.0   0.0         0.0    0.0   0.0             0.0 ␣
 ↪        0.0    0.0          0.0       0.0    0.0          0.0      0.0        ␣
 ↪   0.0     0.0        0.0             0.0               0.0     0.0           ␣
 ↪          0.0       0.0                 0.0          0.0          125.962733␣
 ↪     0
8782208   1307763 2018-10-08   200.0      F  233.14240     Khloe  free    Moore␣
 ↪2018-08-11 20:41:52 2018-10-11 15:58:15  Logged In     NextSong    61487.0 ␣
 ↪Sacramento--Roseville--Arden-Arcade, CA           94.0  Mozilla/5.0 (Windows␣
 ↪NT 6.1; WOW64; rv:31.0) Gecko/20100101 Firefox/31.0          Hands Held High␣
 ↪(Album Version)      Linkin Park    PUT 0 days 00:03:10             190.0    ␣
 ↪125.962733       1.0   0.0         0.0    0.0   0.0              0.0          0.
 ↪0    0.0          0.0       0.0    0.0          0.0      0.0             0.0  ␣
 ↪ 0.0        0.0            0.0              0.0     0.0                       ␣
 ↪ 0.0       0.0                  0.0          0.0          125.962733      0
8782207   1307763 2018-10-08   200.0      F        NaN     Khloe  free    Moore␣
 ↪2018-08-11 20:41:52 2018-10-11 15:55:05  Logged In  Roll Advert    61487.0 ␣
 ↪Sacramento--Roseville--Arden-Arcade, CA           93.0  Mozilla/5.0 (Windows␣
 ↪NT 6.1; WOW64; rv:31.0) Gecko/20100101 Firefox/31.0                           ␣
 ↪          None             None     GET 0 days 00:00:17             17.0    ␣
 ↪125.962733       0.0   0.0         0.0    0.0   0.0              0.0          1.
 ↪0    0.0          0.0       0.0    0.0          0.0      0.0             0.0  ␣
 ↪ 0.0        0.0            0.0              0.0     0.0                       ␣
 ↪ 0.0       0.0                  0.0          0.0          125.962733      0
8782206   1307763 2018-10-08   200.0      F  207.15057     Khloe  free    Moore␣
 ↪2018-08-11 20:41:52 2018-10-11 15:54:48  Logged In     NextSong    61487.0 ␣
 ↪Sacramento--Roseville--Arden-Arcade, CA           92.0  Mozilla/5.0 (Windows␣
 ↪NT 6.1; WOW64; rv:31.0) Gecko/20100101 Firefox/31.0                        ␣
 ↪Holdin' On Together          Phoenix     PUT 0 days 00:03:52             232.0 ␣
 ↪   125.962733       1.0   0.0         0.0    0.0   0.0              0.0       ␣
 ↪ 0.0    0.0          0.0       0.0    0.0          0.0      0.0             0.0␣
 ↪    0.0        0.0            0.0              0.0     0.0                    ␣
 ↪    0.0       0.0                  0.0          0.0          125.962733      0

[3200753 rows x 47 columns]
```

[32]:
```python
print(df_first_weeks['churn'].value_counts(normalize=True))
```

```
churn
0    0.999371
1    0.000629
Name: proportion, dtype: float64
```

6 Training the Churn Model (Rolling Prediction)

[58]:
```python
def rolling_churn_prediction(df, features, label, model=None):
    """
    Perform week-to-week churn predictions using a rolling window approach.

    Parameters:
    - df: Aggregated user activity dataframe with churn labels.
    - features: List of feature column names.
    - label: Target column name (churn).
    - model: Machine learning model (default: Random Forest with␣
↪class_weight='balanced')

    Returns:
    - None (prints classification reports for each week).
    """
    if model is None:
        model = RandomForestClassifier(n_estimators=100, random_state=42,␣
↪class_weight='balanced')  # Handle class imbalance

    weeks = sorted(df['week'].unique())  # Get sorted list of weeks
    f1_scores = []
    week_labels = []

    for i in range(len(weeks) - 1):  # Stop at second-to-last week (since we␣
↪predict next week)
        train_weeks = weeks[:i + 1]  # Train on all past weeks up to the␣
↪current one
        test_week = weeks[i + 1]  # Predict for the next week

        train = df[df['week'].isin(train_weeks)]
        test = df[df['week'] == test_week]

        if test.empty:  # Skip if no data for the next week
            continue

        # Convert categorical features to numeric
        X_train = train[features]
        y_train = train[label]
        X_test = test[features]
        y_test = test[label]

        X_train, X_test = X_train.align(X_test, join="left", axis=1,␣
↪fill_value=0)

        # Impute missing values in X_train and X_test
```

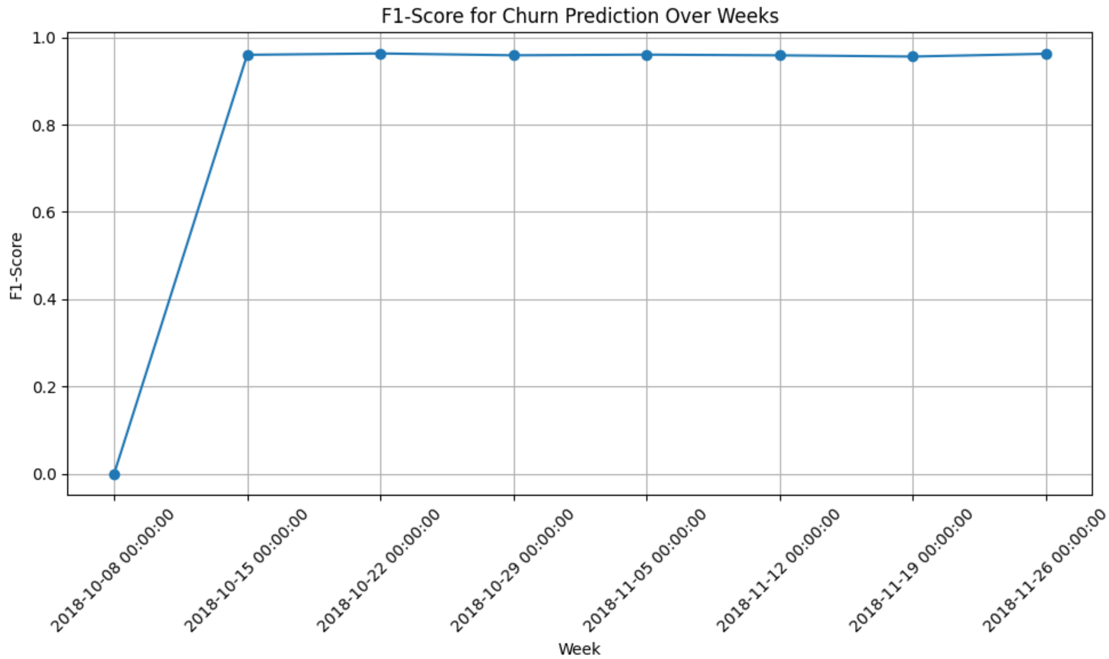
```
        imputer = SimpleImputer(strategy='mean')  # You can use other␣
 ↪strategies like 'median' or 'most_frequent'
        X_train_imputed = imputer.fit_transform(X_train)
        X_test_imputed = imputer.transform(X_test)

        # No SMOTE applied here, directly use the imputed data
        X_train_res, y_train_res = X_train_imputed, y_train

        # Train the model
        model.fit(X_train_res, y_train_res)

        # Predict churn
        y_pred = model.predict(X_test_imputed)

        # Print evaluation results for each week
        print(f"Week {test_week}:")
        print(classification_report(y_test, y_pred, zero_division=0))

        f1 = f1_score(y_test, y_pred, zero_division=0)
        f1_scores.append(f1)
        week_labels.append(str(test_week))

        print("-" * 50)

    # Plot F1-scores
    if f1_scores:
        plt.figure(figsize=(10, 6))
        plt.plot(week_labels, f1_scores, marker='o')
        plt.title('F1-Score for Churn Prediction Over Weeks')
        plt.xlabel('Week')
        plt.ylabel('F1-Score')
        plt.xticks(rotation=45)
        plt.grid(True)
        plt.tight_layout()
        plt.show()
    else:
        print("No evaluations performed (possibly no data after filtering).")
```

[60]:

```
# Fitting the model
model = XGBClassifier(n_estimators=100, random_state=42, base_score=0.5,␣
 ↪scale_pos_weight=10)
features = ['session_count', 'session_duration', 'prev_week_activity',␣
 ↪'dummy_level']
label = 'churn'

rolling_churn_prediction(df = df_all_weeks, features=features, label=label,␣
 ↪model=model)
```

```
Week 2018-10-08 00:00:00:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00   3196811
           1       0.00      0.00      0.00      3942

    accuracy                           1.00   3200753
   macro avg       0.50      0.50      0.50   3200753
weighted avg       1.00      1.00      1.00   3200753

--------------------------------------------------
Week 2018-10-15 00:00:00:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00   3199832
           1       1.00      0.92      0.96      3780

    accuracy                           1.00   3203612
   macro avg       1.00      0.96      0.98   3203612
weighted avg       1.00      1.00      1.00   3203612

--------------------------------------------------
Week 2018-10-22 00:00:00:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00   3142430
           1       1.00      0.93      0.96      3597

    accuracy                           1.00   3146027
   macro avg       1.00      0.96      0.98   3146027
weighted avg       1.00      1.00      1.00   3146027

--------------------------------------------------
Week 2018-10-29 00:00:00:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00   3047176
           1       1.00      0.92      0.96      3610

    accuracy                           1.00   3050786
   macro avg       1.00      0.96      0.98   3050786
weighted avg       1.00      1.00      1.00   3050786

--------------------------------------------------
Week 2018-11-05 00:00:00:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00   2955396
```

```
           1       1.00      0.92      0.96      3477

    accuracy                           1.00   2958873
   macro avg       1.00      0.96      0.98   2958873
weighted avg       1.00      1.00      1.00   2958873

--------------------------------------------------
Week 2018-11-12 00:00:00:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00   2909850
           1       1.00      0.92      0.96      3387

    accuracy                           1.00   2913237
   macro avg       1.00      0.96      0.98   2913237
weighted avg       1.00      1.00      1.00   2913237

--------------------------------------------------
Week 2018-11-19 00:00:00:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00   2589723
           1       1.00      0.92      0.96      3623

    accuracy                           1.00   2593346
   macro avg       1.00      0.96      0.98   2593346
weighted avg       1.00      1.00      1.00   2593346

--------------------------------------------------
Week 2018-11-26 00:00:00:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00   2204483
           1       1.00      0.93      0.96      3526

    accuracy                           1.00   2208009
   macro avg       1.00      0.96      0.98   2208009
weighted avg       1.00      1.00      1.00   2208009

--------------------------------------------------
```


F1-Score for Churn Prediction Over Weeks
1.0
0.8
0.6
0.4
0.2
0.0
F1-Score
2018-10-08 00:00:00
2018-10-15 00:00:00
2018-10-22 00:00:00
2018-10-29 00:00:00
2018-11-05 00:00:00
2018-11-12 00:00:00
2018-11-19 00:00:00
2018-11-26 00:00:00
Week